

Hydrogen Membrane engineering manual

Track hydrogen recovery, permeate purity, feed pressure and the feed-to-non-permeate flow ratio.

Run the Python model

Open `membrane_analyzer_engine.py` in Python IDLE. Select Run, then Run Module (F5). Python 3.10 or newer is required; no extra packages or other files are needed.

The built-in example writes a results JSON file beside the Python file. Edit SETTINGS near the top and run again to change assumptions. To use your own measurements, run the file with an input JSON filename. Inputs use the fields and units below.

The input envelope contains `schemaVersion`, `model`, `source`, `timezone`, `config` and `measurements`. Each measurement contains `timestamp`, `values` and optional `quality`. Use ISO timestamps with an offset. Bad, stale and missing signals remain unavailable. The AVEVA mapping is provisional; a live connection has not been verified.

Inputs and units

Model inputs • 5 rows		
INPUT	UNIT	USE
Timestamp; feed and permeate flow	date/time; Nm³/h	Hydrogen recovery and flow balance.
Non-permeate flow	Nm³/h	Flow ratio; derived as feed minus permeate if absent.
Feed and permeate H ₂ content	vol%	Separate online analyzer and laboratory readings.
Feed pressure	kPag	Compare with the design pressure.
Design values and condition limits	Configuration	Recovery, purity, flow ratio and pressure references.

Calculations

HYDROGEN RECOVERY

Recovery = $100 \times (Q_{\text{permeate}} \times y_{\text{H}_2, \text{permeate}}) / (Q_{\text{feed}} \times y_{\text{H}_2, \text{feed}})$

Online and lab cases use their respective compositions. All flows must share the same reference conditions.

PURITY AND FLOW RATIO

Purity = permeate H₂%; ratio = $Q_{\text{feed}} / Q_{\text{non-permeate}}$

Purity is an input measurement, not a predicted membrane composition.

PRESSURE DEVIATION

$$\text{Deviation} = 100 \times (P_{\text{feed}} - P_{\text{design}}) / P_{\text{design}}$$

Gauge-pressure comparison; the alert checks absolute percentage deviation.

Useful trends

- Recovery and purity together to show the recovery/purity trade-off.
- Feed H₂ and feed pressure alongside recovery changes.
- Feed, permeate and non-permeate flows; total/non-permeate ratio against its limit.
- Online and lab results to check analyzer disagreement at matched sample times.

Assumptions and limits

- If the whole dataset lacks online feed H₂, a labelled synthetic trace is interpolated from lab samples. It is not an online measurement. Individual missing online values can use the lab value.
- Lab recovery still uses measured flow inputs; it is not independent verification of all instruments. Recovery above 100% or a negative flow balance requires input review.
- This is a material-balance monitor, not a membrane transport, remaining-life or failure-probability model.

Measured values come from instruments or samples. Calculated values use those inputs and the configured assumptions. Design values are references. Simulated values demonstrate the model and are not plant measurements. Missing results remain unavailable.

Sources

- Supplied Membrane Performance workbook: hydrogen balance, purity, pressure and flow-ratio requirements